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WO-100-04 — Remake flanged joint 2"-P300-LS-3007/A2 on 2"-P300-LS-3007

Area	A100 Tank farm	**Task**	GJ Flanged joint remake
Priority	1 (safety)	**Issued by**	Facility owner (demonstration)
Primary asset	T-101 — T-101 tank 12' dia × 22' high — E 25'-0" N 26'-6" EL 100'-6"		
Equipment	T-101	**Lines**	2"-P300-LS-3007
Isometrics	ISO-LS-3007 (owner's drawings, attached)		
Permits	General work permit		

Reason

Steam leak at the tank heating-coil inlet flange.

Scope of work

1. Isolate per the isolation list, drain and vent 2"-P300-LS-3007.
2. Break the joint at E 22'-0" N -12" EL 123'-9-1/4" between FLANGE WELD-NECK 2" P300 RAISED FACE and VALVE, GATE, WEDGE, OS&Y 2" P300.
3. Inspect both flange faces (SW-NDE-5).
4. Fit a new gasket and new studs; tighten to the stated preload in three passes, cross pattern (SW-WO-5).
5. Remove blinds; leak test on start-up (SW-TEST-2).

Affected items: 2"-P300-LS-3007/A2, 2"-P300-LS-3007/A1, 2"-P300-LS-3007/A3

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-LS-3007	2"-P300-LS-3007	2"-P300-LS-3007	1	agrees with the model

ISO-LS-3007 (PDF): ../isometrics-imported/ISO-LS-3007.pdf — the sheets that hold the work follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-101	equipment	T-101 tank 12' dia × 22' high		E 25'-0" N 26'-6" EL 100'-6"
2"-P300-LS-3007/A2	item	GASKET 2" P300 COMPRESSED SHEET	2"-P300-LS-3007	E 22'-0" N -12" EL 123'-9-1/4"
2"-P300-LS-3007/A1	item	FLANGE WELD-NECK 2" P300 RAISED FACE	2"-P300-LS-3007	E 22'-0" N -12" EL 123'-7-1/8"
2"-P300-LS-3007/A3	item	VALVE, GATE, WEDGE, OS&Y 2" P300	2"-P300-LS-3007	E 22'-0" N -12" EL 124'-2-11/16"
6"-P300-LS-2003/T:2"-P300-LS-3007	item	TEE REDUCING 6" × 2" P300	6"-P300-LS-2003	E 22'-0" N -12" EL 123'-0"
2"-P300-LS-3007/B2	item	VALVE, GATE, WEDGE, OS&Y 2" P300	2"-P300-LS-3007	E 18'-5/8" N 26'-6" EL 102'-6"
2"-P300-LS-3007/B5	item	FLANGE WELD-NECK 2" P300 RAISED FACE	2"-P300-LS-3007	E 17'-5-1/16" N 26'-6" EL 102'-6"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	6"-P300-LS-2003/T:2"-P300-LS-3007	6"-P300-LS-2003	No valve on this side within the line: isolate or depressure the parent line 6"-P300-LS-2003 at its tee under the shutdown isolation			
2	valve	2"-P300-LS-3007/B2	2"-P300-LS-3007	Close and lock VALVE, GATE, WEDGE, OS&Y 2" P300 (2"-P300-LS-3007/B2)	E 18'-5/8" N 26'-6" EL 102'-6"		
3	blind	2"-P300-LS-3007/B5	2"-P300-LS-3007	Fit a line blind between 2"-P300-LS-3007/B2 and FLANGE WELD-NECK 2" P300 RAISED FACE (2"-P300-LS-3007/B5), work side	E 17'-7-1/8" N 26'-6" EL 102'-6"		

Parts

Description	Qty	Replaces
GASKET 2" P300 COMPRESSED SHEET	1	2"-P300-LS-3007/A2
STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300	1	2"-P300-LS-3007/A4

Inspections

Id	Inspection	Criteria	Basis
WO-100-04-I1	Flange faces before the new gasket goes in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
2"-P300-LS-3007/A2	4 × 0.625" studs (2"-P300-LS-3007/A4)	13,806	3	

Test — Service leak test

Description	The remade joint checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

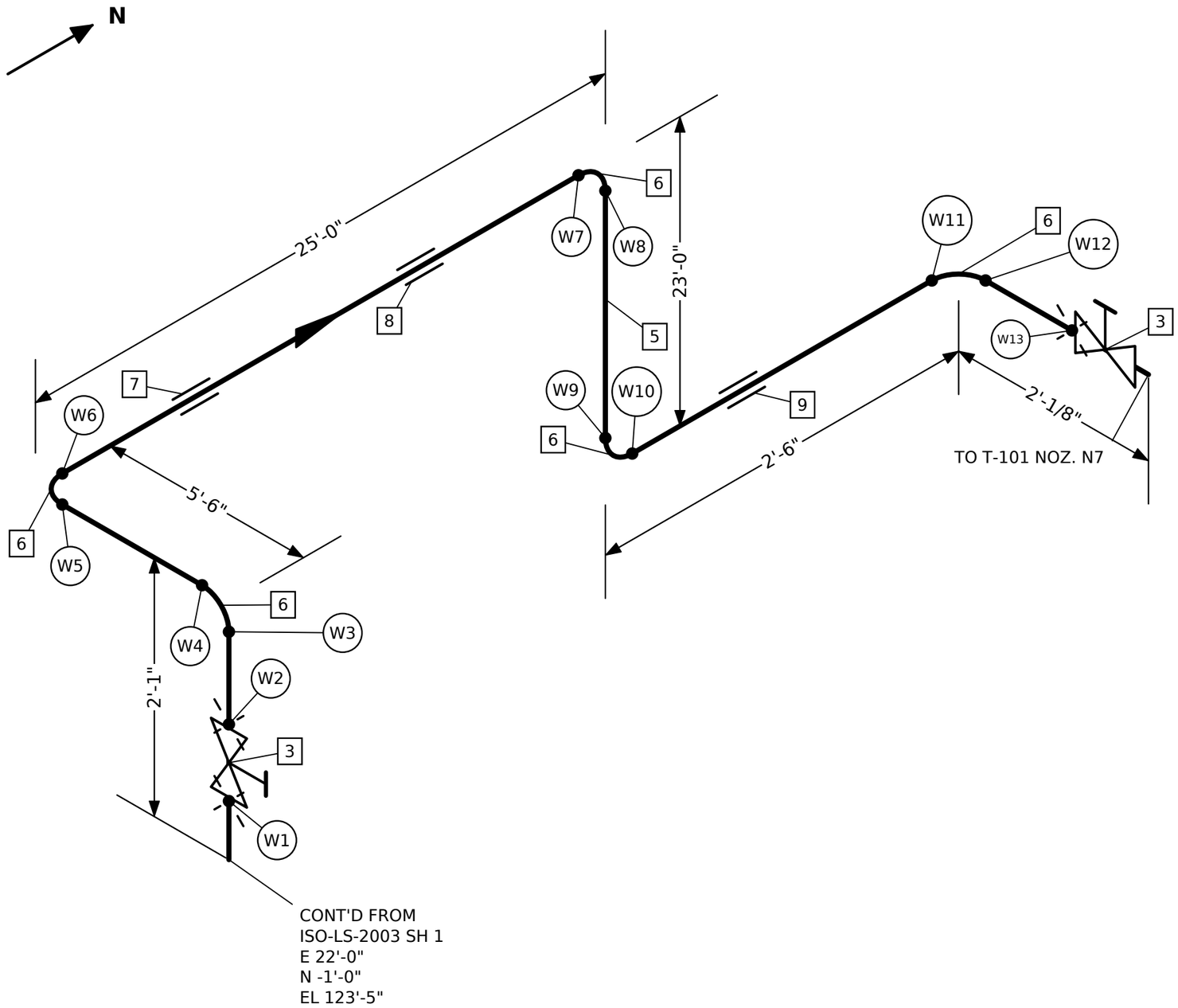
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Joint broken; both flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New gasket fitted; studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated	Bolt-up record	W	W	—				
4	Service leak test on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
5	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
6	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
2"-P300-LS-3007/A2	affected	ISO-LS-3007	1	
2"-P300-LS-3007/A1	affected	ISO-LS-3007	1	
2"-P300-LS-3007/A3	affected	ISO-LS-3007	1	
6"-P300-LS-2003/T:2"-P300-LS-3007	isolation: equipment	ISO-LS-2003	1	
2"-P300-LS-3007/B2	isolation: valve	ISO-LS-3007	1	
2"-P300-LS-3007/B5	isolation: blind	ISO-LS-3007	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	2"	FLANGE WELD-NECK 2" P300 RAISED FACE
2	4	2"	GASKET 2" P300 COMPRESSED SHEET
3	2	2"	VALVE, GATE, WEDGE, OS&Y 2" P300
4	4	2"	STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300
5	54'-5 1/4"	2"	PIPE 2" P300 BEVELLED ENDS
6	5	2"	ELBOW 90° R = 5/3 OD 2" P300
7	1	2"	HANGER FOR 2" LINE (HANGER-2x87.0)
8	1	2"	HANGER FOR 2" LINE (HANGER-2x77.0)
9	1	2"	TRUNNION FOR 2" LINE (TRUNNION-2x29.0)

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. CHECK ORIENTATION BEFORE WELDING

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

LOW-PRESSURE STEAM

TEST PRESS.	INSULATION	PAINT
-	YES	-

PWHT	NDE	FLUID
-	-	LS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P300-LS-3007	S300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-LS-3007	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

Notices

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